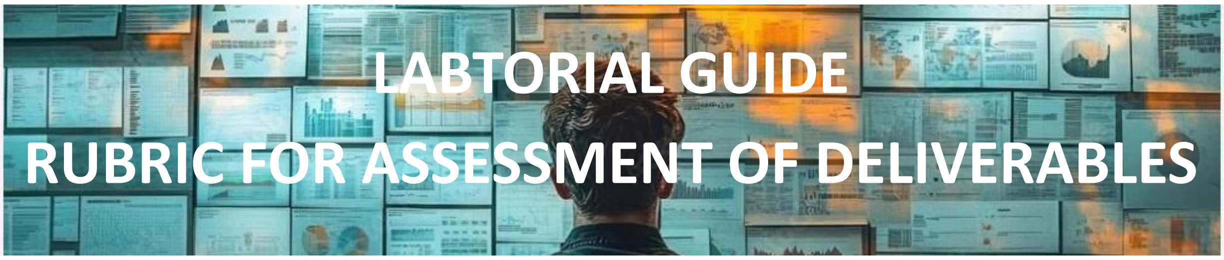

LABTORIAL GUIDE
RUBRIC FOR ASSESSMENT OF DELIVERABLES

*"*Buckle your seatbelt Dorothy, 'cause Kansas … is going bye-bye!*" (The Matrix, 1999)*

Table of contents

Purpose

This document belongs to ICT2113 module of the SIT cluster of the same acronym (ICT). Its purpose is to provide explain, clarify, and guide all conduct of assessing the module's required and assessable deliverables and performances by the students, a.k.a., their assessable items. The key information articulated and presented therein and to be evaluated, are partitioned into designated aspects of the assessable items. The goodness of the articulation and presentation of the information in said aspects are scored via a prescribed set of degrees of quality. Finally, each aspect is assigned a weighting, i.e., its relative magnitudinal significance to the final score for each aspect of those items. By sharing this process to the students, the assessment transparency is ensured.

Fundamentals

This module has three families of *Assessable items*, namely team and individual deliverables into the LMS, and individual performance in staged presentation to audience comprising class peers (see Figures 4a & b of the Labtorial Guide).

Every assessable item is partitioned into a fixed set of inherent *Aspects*. For deliverables, these aspects include content, styling, and writing quality. For performance, they include delivery style and level of knowledge. Any Assessable item that is posted or carried out by the team or individual, is designated as an Attempt.

Every Aspect in the Attempt is evaluated for its degree of quality (DoQ) or "goodness". Said evaluation is qualitative, not quantitative or numeric. This is in keeping with known psychology of measurement (see next section "Enumeration…"). (Note there is never an evaluation of the Assessable item as a whole; only its partitions are evaluated.) Each DoQ is assigned a quantitative Score, that is in turn adjusted into a weighted Score via its multiplication with the Weighting factor prescribed for that Aspect. The total of the weighted Scores renders the Mark for the Attempt. Finally, the evaluator will review the collection of Marks for consistency across every Attempt of the Assessable item, and tweak as necessary.

At the most recent count before add-drop period, there are 21 teams and 108 individuals. The following are standard rules underpinning all submissions:

1. If two or more students in the same group post an Attempt of the Assessable item, only that from the PoC will be evaluated. If the POC is not amongst the student posters, then the earliest Attempt by those non-PoC posters will be evaluated.
2. Several Attempts of an Assessable item is permitted, but only the most recent will be evaluated (except in the case of rule 1).
3. Any and all Attempts that are posted after the deadline of Friday 1800 hrs will not be evaluated. There is no appeal.

Enumeration: Non-numeric vs Numeric

The previous section "Fundamentals" explains that all Attempts are evaluated qualitatively by the evaluator and then auto-translated into a corresponding Score. In other words, the evaluator does not just do the numeric evaluation and hence bypass the extra inceptive effort of doing the qualitative evaluation. Why is this the best approach for doing the evaluation?

This comes from the agreed findings of the science of psychological measurement called Psychometrics. Simply put, humans are better at categorical judgment than magnitudinal estimation, especially when the range of the magnitude scale is large, like 0 to 100. Here is a brief layman's explanation.

1. Humans are naturally wired to categorize qualitatively things in all sorts of ways. This is known as 'cognitive ease of qualitative labeling'. In other words, qualitative labels like poor, average, good, excellent, etc, reside inside the mind as stable anchors that can be repeatedly recalled and applied consistently across many items being judged. For example, one could probably judge a hundred documents for poor, average, good, excellent in a reasonably accurate and consistent effort. But if one had to assign a magnitude from a simple scale like 0..9, 10..24, 25..39, and 40..50, the same accuracy and consistency would not be derived. Additionally, two different people are much more likely to agree in how they evaluate those documents as being poor, average, good, or excellent, than they are to assign exactly the same numeric scores.
2. Some believe that the larger numeric scales are more precise, e.g., a value in the range 0..100 is considered more precise than in 1..10. This is true only if one is reading it. But the notion of precision vanishes when one is asked to score something on any scale, but especially large scales. The problem is not just the absence of qualitative guidance, but a problem of granularity, that in turn requires more cognitive load. On a 100-point scale, the difference between a 60 and a 70 is more subjective than between 6.0 and 7.0. Plus, shifts in the evaluator's mood or fatigue can easily influence the magnitude estimation results: a "60" given on Monday might be an "80" given on Friday.

So, as it turns out, the best approach is for humans to do a qualitative evaluation first, and then to follow that up with a translator for generate the numeric equivalents, especially when many Instances comprised of many Attempts have to be assessed.

Assessable items & their Aspects

Team deliverable

1. **Project specification (PS)** – 10 Aspects.
    - **Project description** – name; context; and, goal.
    - **Reqrs tree** – contextually relevant & meaningful (CRaM) organization of reqrs by granularity; and, its congruous application to total set of reqrs clauses.
    - **Biz ops reqrs** – assumptions; company constraints & rules; external biz systems; govt & international regs; primary biz systems consistent with profile; and, judiciously assigned precedence indices.

- **IT tech reqrs** – company platform; database; and, networking & APIs.
- **Codification** – for reqrs only.
- **Financials** – payment schedule (by percentage, not enumeration); and, project estimated costing. (Highlight its redaction of costs in released version.)
- **Scheduling** – milestones for delivery; and, progress reporting.
- **Sign-off** (assume validated and ready for approval from all stakeholders) – authenticator signature; and, designated approvers (not yet signed).
- **Reqrs modeling** – appropriate for customer per availability constraints.
- **Addenda** – purpose-appropriate format & structure (F&S); and, content that is CRaM.

2. **Analysing report** – 8 Aspects.

- **Executive summary** – correct and is CRaM content.
- **BCIC** – creative content that is CRaM for each method.
- **IRC** – creative content for team IRC that is CRaM.
- **Repository** – actual link for team repository (public); and, site org that is CRaM.
- **Account manager** – creative content that is CRaM.
- **Risks analysis** – creative content that is CRaM.
- **Reqrs modeling** – necessary for vendor to demo to stakeholders.
- **Sign-off** (assume ready for submission to vendor C staff) – signature of lead of analysis team.

3. **Software requirements specification (SRS)** – 13 Aspects

- **Project description** – summary of main features; assumptions; and implementation constraints.
- **Reqrs tree** (different nodes than for Eliciting) – CRaM organization of reqrs by granularity; and, its congruous application to total set of reqrs clauses.
- **Template** – name of owning official standard or unofficial source, e.g., web site.
- **Reqrs** – export from repository and fit to your Reqrs tree with sub-headings dictated by your team's chosen template.
- **Latest tech** – operating environment including platform.
- **Quality dev model** – creative content for five CRaM quality feature names & descriptions of what each covers or measures.

- **Codification** – mandatory for reqrs, tech, quality, iRTM, & change mitigation (follow template if it has more, but not if it has less than mandatory).
- **iRTM** – correct content as duplicates, paraphrasing, or minimizing of reqrs clauses in PS and (this) SRS.
- **Change mitigation** – derived from LS info, or creative content that correctly specifies change mitigation strategies.
- **Addenda** – correct F&S and content that is CRaM.
- **Reqs modeling** – all in PS & repository are brought into the SRS, either as a duplicate or with corrections as necessary; and 'figure callout' or 'in-text reference' to each model.
- **Sign-off** – vendor authenticator sign-off; and, designated approvers in Customer and Vendor (without signatures).
- **References** – short list of the most significant and impactful written documents or recorded discussions either minuted or digital.

4. **Validating combined report (QC & QA)** – 5 Aspects each.

- **Executive summary for QC** – correct and CRaM content.
- **Errors detected by QC** – list of Authorship error instances; list of Omission error instances; and, list of Contradiction error instances.
- **AI review of QC method conduct & detections** – show copy of prompts; show copy of AI responses.
- **Final findings of QC team** – correct and CRaM content.
- **Sign-off for QC** (assume ready for submission to vendor C staff) – signature of lead for QC audit team.
- **Executive summary for QA** – correct and CRaM content.
- **Errors detected by QA** – list of Process non-compliance error instances.
- **AI review of QA method conduct & detections** – show copy of prompts; show copy of AI responses.
- **Final findings of QA team** – correct and CRaM content.
- **Sign-off for QA** (assume ready for submission to vendor C staff) – signature of QA audit lead.

5. **Post-Agile combined report (User story & UseCase2.0)** – 7 Aspects.
    - **Executive summary** – correct and CRaM content.
    - **User stories** – five clauses, each written in as per one effective writing convention.
    - **Conversations or interviews** – five with content creative and CRaM (one for each User story).
    - **UseCase graph** – derived from PD, and highlights UseCase to be used for Story and Slices. Also, each model must have 'figure callout' or 'in-text reference' to it.
    - **Stories** – five whose contents are CRaM. Also, each Story must have codified reference to it.
    - **Slices** – two derived from Stories, one being multi-Story, that are CRaM. Also, each Slice must have codified reference to it.
    - **Sign-off** (assume ready for submission to customer C staff) – signature of lead of analysis team.

6. **Analysing slide deck** – It is posted along with report.

7. **QC report slide deck** – It is posted along with report.

8. **QA report slide deck** – It is posted along with report.

<u>Notes for above deliverables 1-5</u>

A. The following Aspects are required, and must be CRaM for each particular document:
    - **Progenitor** – correct profile; correct artifacts applied consistently throughout; and, covering letter.
    - **Org & styling** – key authoring participants; page numbering (all main doc pages); purpose-appropriate F&S; and, ToC.
    - **Maximum pages** (not including addenda) – met within one additional page.
    - **Declarations** (assume company policy) – one for AI; and, one for plagiarism.

B. In Validating report only, above single Projector, Org & styling, and Maximum pages, are replaced by dual Aspects as follows:
    - **Progenitor for QC –** same characteristics as per item "A." above.
    - **Progenitor for QA** – same characteristics as per item "A." above.
    - **Org & styling for QC** – same characteristics as per item "A." above.

- **Org & styling for QA** – same characteristics as per item “A.” above.
- **Maximum pages for QC** – same characteristics as per item “A.” above.
- **Maximum pages for QC** – same characteristics as per item “A.” above.

Individual deliverable

9. **Midterm quiz.**
    - **Content** – compares favourably with model answer.
    - **No AI or plagiarism** – level of similarity or direct copying must be under 50%.
    - **Maximum words** – met within five additional words.

10. **Final quiz** – same Aspects as per Midterm quiz.

Individual performance

11. **Presenting part of Analysing report.**
    - **Attention grabbing** – eye contact with members of the audience; walk away from the podium; and, suitably (not overly) animated so as to liven things up.
    - **No detracting mannerisms** – no: droning; reading slides; being fidgety; or using filler words.
    - **Knowledgeable** – doesn’t refer to handheld info prompting instrument for majority of his/her presentation, or to answer questions from the audience.

12. **Presenting part of QC-part of combined report** – same Aspects as per Midterm quiz.

13. **Presenting of QA-part of combined report** – same Aspects as per Midterm quiz.

Notes for all deliverables 1-13

C. In the event one or more extra Aspects are in any document, being as follows below, and not listed above, one said ‘extra’ may substitute for one ‘missing standard Aspect’. If none missing, then it may be added as a bonus.

- **Xtra** – any unique artifact, aspect, effort level, or provision, that belongs to SWE and is CRaM. Note, weighting is either PK or DBK, at discretion of evaluator.

Qualitative evaluation

DoQ (or goodness)

As stated earlier, the Attempts of the separate Aspects of the Assessable item are evaluated, the evaluation is qualitative not quantitative, and the evaluation assigns DoQ to each Attempt. From the previous section, every Aspect has characteristics that can be observed and/or measured for DoQ or goodness. As there are several characteristics for most Aspects, the DoQ of its Attempt covers all the characteristics. If any characteristic is missing or poorly articulated, the DoQ will be less than that if all characteristics are present and well-articulated. Ergo, the DoQ is not a zero-sum assignment; instead, it is more of a normalization. Delineation of the DoQ assignment is based on a seven-point Likert scale.

N-il. The assessable item is not posted/uploaded into the LMS, or the expected presentation is not carried out.

V-ery poor. The specified deliverable is uploaded, or the expected presentation is carried out. The Attempt either covers one/two of the details listed for that Aspect, and the expected/required info or conduct pertaining to it/them detail is totally flawed'; it's as if the progenitor hasn't learned anything.

Po-or. The specified deliverable is uploaded, or the expected presentation is carried out. The Attempt either covers one/two of the details listed for that Aspect, and the expected/required info or conduct pertaining to it/them detail is mostly flawed; it's as if the progenitor is doesn't know the real answer/conduct and can only come up with a guess.

O-k. The specified deliverable is uploaded, or the expected presentation is carried out. The Attempt covers most of the details listed for that Aspect, but the expected/required info or conduct pertaining to that detail is mostly flawed; it's as if the progenitor knows the bare minimum about the answer/conduct.

G-reat. The specified deliverable is uploaded, or the expected presentation is carried out. The Attempt either covers all/most of the details listed for that Aspect, and the expected/required info or conduct pertaining to that detail is a mix of flawless and flawed; it's as if the progenitor has learned some of the answer/conduct well.

E-xcellent. The specified deliverable is uploaded, or the expected presentation is carried out. The Attempt covers all of the details, and the expected/required info or conduct pertaining to those details are mostly flawless; it's as if the progenitor has learned most of the answer/conduct well.

Pe-rfect. The specified deliverable is uploaded, or the expected presentation is carried out. The Attempt covers all of the details listed for that Aspect, and the expected/required info or conduct pertaining to each and every detail is utterly flawless; it's as if the progenitor has mastered every bit of the answer/conduct and may well even gone further into learning material outside of the module.

Weighting

Weighting is an enumeration of the relative importance of a kind of Aspect compared to the other kinds of Aspects that comprise an Assessable item. It is qualitative and prescribed to every kind of Aspect before the Attempt undergoes DoQ evaluation. There are three kinds of weighting, and these are assigned according to the evaluators best understanding of the learning significance of each Aspect.

Procedural knowledge (PK). This weighting applies to the Aspects of the instruments that are used to deliver in an industry-standard manner, the academic message that the assessment activity dictates. It is important from an administrative perspective, and is often is the first of a series of evaluations conducted on the Assessable item. It is also the easiest to evaluate as its presence, completeness, and correctness are mostly evident by purely being observed. Therefore, its weighting is the lowest.

Declarative basic knowledge (DBK). This weighting applies to the characteristics that embody and exhibit the basic, foundational & primary characteristics of the academic message. It is important from a contextual perspective, and establishes the academic foundation upon which the definitive part of the message can be evaluated. It is evaluated by comparing the compatibility, soundness & veracity of the Attempt against the expected/required characteristics of the Aspects. Its weighting is in between the three weightings.

Declarative definitive knowledge (DDK). This weighting applies to the characteristics that embody and exhibit the definitive, essential & threshold characteristics of the academic message. It is important from a provisional perspective, and provides the core information and underpinning purpose that its creation is meant to offer. It is be evaluated by comparing the compatibility, soundness & veracity of the Attempt against the expected/required characteristics of the Aspects. Its weighting is the highest.

Assignment of weighting to Aspects

1. **Project specification (PS).**

Weighting	Aspect
PK	Progenitor
PK	Org & styling
PK	Max pages
PK	Declarations
PK	Addenda
DBK	Project description
DBK	Reqrs tree

Weighting	Aspect
DDK	Biz ops reqrs
DDK	IT tech reqrs
DDK	Financials
DDK	Scheduling
DDK	Reqrs modeling
DBK	Codification
DBK	Sign-off

2. **Analysing report.**

Weighting	Aspect
PK	Progenitor
PK	Org & styling
PK	Max pages
PK	Declarations
PK	Sign-off
DBK	Executive summary

Weighting	Aspect
DDK	BCIC
DDK	IRC
DDK	Account manager
DDK	Risks analysis
DDK	Reqrs modeling
DBK	Repository

3. **Software requirements specification (SRS).**

Weighting	Aspect
PK	Progenitor
PK	Org & styling
PK	Max pages
PK	Declarations
PK	Addenda
DBK	Project description
DBK	Reqrs tree
DBK	Template
DBK	References

Weighting	Aspect
DDK	Reqrs
DDK	Latest tech
DDK	Quality dev model
DDK	iRTM
DDK	Change mitigation
DBK	Reqrs modeling
DBK	Codification
DBK	Sign-off

4. **Validating combined report (QC & QA).**

Weighting	Aspect
PK	Progenitor for QC
PK	Org & styling for QC
PK	Max pages for QC
PK	Sign-off for QC
DBK	Executive summary
DDK	Errors detected by QC
DDK	Final findings of QA

Weighting	Aspect
PK	Progenitor for QA
PK	Org & styling for QA
PK	Max pages for QA
PK	Sign-off for QA
DBK	Executive summary
DDK	Errors detected by QA
DDK	Final findings of QA

DBK	AI review of QC mthd

DBK	AI review of QC mthd

5. **Post-Agile combined report (User story & UseCase2.0).**

Weighting	Aspect
PK	Progenitor
PK	Org & styling
PK	Max pages
PK	Declarations
PK	Sign-off
DBK	Executive summary

Weighting	Aspect
DDK	User stories
DDK	Conversations/Intrvw
DBK	UseCase graph
DDK	Stories
DDK	Slices

6. **Analysing slide deck**. No weighting; 100% score.

7. **QC slide deck**. No weighting; 100% score.

8. **QA slide deck**. No weighting; 100% score.

9. **Midterm quiz.**

Weighting	Aspect
PK	Max words
PK	No AI or plagiarism

Weighting	Aspect
DDK	Content

10. **Final quiz.** Same as for Midterm quiz.

11. **Presenting part of Analysing report**.

Weighting	Aspect
PK	Attention grabbing
PK	No detracting manner

Weighting	Aspect
DDK	Knowledgeable

12. **Presenting part of QC-part of combined report.** Same as for Analysing report.

13. **Presenting of QA-part of combined report.** Same as for Analysing report.

Qualitative evaluation

Score for DoQ enumerations

There are seven DoQ enumerations; these are the corresponding seven Scores.

DoQ	N	V	Po	O	G	E	Pe
Score	0	0.2	0.4	0.5	0.6	0.75	0.85

Weighting factor

Total of weight must be 100.

Weighting	PK	DBK	DDK
Weight %	20	0.35	0.45

Epilogue

Consider the posting of this instruction to be a soft launch. Although great care and thought has gone into this document, it is unlikely that it is without incompleteness and inconsistency. So, the author is grateful for any problems with this instruction to be brought to his attention.

Additionally, the teacher will take the opportunity to check the suitability of these instructions at the start of each of the practice labs. This is a good forum for the students to seek any clarifications that they require.

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